

Civic Expansion and Targeted Violence under Armed Governance: Evidence from Colombia*

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Abstract

Do grassroots civic institutions protect local leaders from violence, or expose them? Using Colombia's *Juntas de Acción Comunal* (JACs) and a novel 65-year administrative panel, I exploit within-municipality variation in annual JAC registrations in a continuous-treatment difference-in-differences design. On the full panel, civic expansion raises subsequent targeted killings of social leaders: the average causal effect is positive and significant, on the order of one additional killing for every ninety new JACs. This effect is concentrated in conflict municipalities and is neutral or protective where armed groups are absent; it is strongest where armed governance is durable, a descriptive gradient identified under stronger assumptions than the baseline. A second finding bounds the threat: killings do not reduce subsequent JAC registrations at any horizon, so targeting does not suppress the formation of new boards, though registration is an imperfect proxy for whether they keep functioning. A leader is targeted for their role in the network, not as an individual, so collective protection is the relevant policy lever.

Keywords: armed governance; selective violence; civic institutions; social leaders; collective action; Colombia

JEL Codes: D71; D74; O17; O54; P16

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1 Introduction

Since the 2016 peace accord between the Colombian government and the Revolutionary Armed Forces of Colombia (FARC)¹, the killing of local civic leaders in Colombia has reached record levels. The *Defensoría del Pueblo* documented over 1,500 killings of social leaders and human rights defenders between January 2016 and mid-2025, among the highest rates in the country’s recorded history (Defensoría del Pueblo, 2025). In the administrative records assembled for this paper, drawn from the *Centro de Memoria Histórica* and spanning 1960–2024, roughly 54 % of recorded victims are *líderes comunitarios*, the category that corresponds most directly to the presidents and board members of Colombia’s *Juntas de Acción Comunal* (JACs), the elected, legally registered neighborhood boards that form the country’s most granular layer of civic life. The most visible and institutionalized local leaders, in other words, are being killed disproportionately, and in precisely the areas where armed governance reorganized after the FARC’s demobilization (Prem et al., 2022).

This is a puzzle, because the prior literature points the other way. A large body of work holds that civic organization *protects*: community institutions build collective monitoring capacity and raise the social cost of predation (Ostrom, 1990; Putnam, 2000), and on the Colombian case specifically Kaplan (2017) documents a robust protective association between junta density and homicide rates. Yet here civic leaders are being hunted, and the killings have intensified precisely as the civic network has expanded. Under what conditions, then, does the expansion of grassroots civic institutions raise, rather than lower, the risk of violence against local leaders? The answer, this paper argues, depends not on whether armed groups are present, but on how durably they govern. Where armed actors pursue durable territorial control, each new JAC densifies the local infrastructure of civic representation and creates named, publicly accountable figures whose organizational salience poses a collective challenge to armed authority. Where armed presence is episodic or absent, the same expansion is protective or inconsequential.

The mechanism runs through institutional visibility. Created by Law 19 of 1958 (Función Pública de Colombia, 1958) in the aftermath of *La Violencia*, the JACs active today are voluntary neighborhood boards that elect a president, sign agreements with municipal governments, execute small public works, and mediate between residents and the state. Each registration thus creates named, publicly listed figures, president, board members, official signatories, who attend municipal planning councils and are identifiable to any organization monitoring local power. This raises the organizational salience of local leaders as a class. The resulting exposure channel produces a positive average effect on targeted killings across the panel, and its strength varies with the strategic horizon of the armed actor: groups with episodic or predatory presence have little strategic interest in eliminating institutional competitors, while groups pursuing durable territorial governance

¹FARC’s origins date back to the early 1950s (InSightCrime, 2023). The group officially adopted the name *Fuerzas Armadas Revolucionarias de Colombia* in 1966, presenting itself as a Marxist-Leninist guerrilla movement advocating land reform and political inclusion. In the early 1980s, the group added the suffix -EP (for *Ejército del Pueblo*, people’s army) to underline its self-identification as a structured, autonomous insurgent force with military and political objectives going beyond traditional guerrilla warfare (Tawse-Smith, 2008). For simplicity’s sake, I will be using the term FARC in the remainder of the article.

internalize the long-run cost of a rival civic institution and face correspondingly strong incentives to act. Civic expansion itself generates the threat this paper documents; governance durability determines where it concentrates.

To test this argument, I exploit a novel administrative panel spanning 1960–2024, constructed from Ministry of Interior registration records documenting the founding date of every JAC in Colombia: 48,767 registrations across 1,124 municipalities. The treatment is the annual flow of newly registered JACs per 10,000 inhabitants. The outcome is targeted killings of social leaders from the *Centro de Memoria Histórica*. Because JAC creation is staggered, continuous, and non-absorbing, I implement the continuous-treatment event-study estimator of de Chaisemartin et al. (2025), which accommodates heterogeneous treatment intensity and identifies effects under parallel trends. Identification rests on within-municipality variation in the timing of JAC creation, supported by evidence that early JAC formation was driven by administrative distance and rural population density rather than by conflict history, and by timing diagnostics that rule out anticipation and reverse causality in both directions.

The dynamic DiD yields a positive and statistically significant average total effect on targeted killings ($\hat{\beta} = 0.00314$, 95% CI [0.0012, 0.0051], placebo $p = 0.338$), an 11.5% increase over the baseline mean in conflict municipality-years at the mean treatment intensity; it is robust across alternative outcome transformations (all significant at the 1% level) and to excluding the post-2016 administratively contaminated zones (Appendix A). This is the first quantitative evidence, from within-municipality variation in a 65-year administrative panel, that civic expansion can causally raise the risk of targeted violence against local leaders. The average conceals two opposite dynamics: where no armed group is documented, the estimate is negative or zero, consistent with the protective role of civic organization documented in prior work; within conflict municipalities, the JAC–violence elasticity rises with the durability of armed governance. A between estimator sharpens this gradient, the established–sporadic differential being positive and significant in the strictly predetermined post-2016 window ($p = 0.029$), which points to the durability of armed control, not the mere incidence of violence, as what activates targeting. The three conflict categories do not form a clean monotone gradient, but entered continuously governance durability does: each additional year of documented armed presence over 1993–2015 raises the elasticity of killings to civic expansion (Table 8). This heterogeneity rests on stronger assumptions than the baseline DiD, and I read it as suggestive support for the governance-durability mechanism, not as an independent causal test. The effect is not a mechanical artifact of JAC-affiliated victims: excluding *líderes comunitarios*, the category that maps most directly to JAC office-holders, halves it but leaves it positive and significant, consistent with civic expansion raising the salience of local leaders as a class. A second, independent finding bounds the threat. Targeted killing does not suppress the registration of new boards at any horizon: a leader is targeted for their role in the civic network, not as an individual, so removing one leaves the network’s institutional pull intact and registration continues, even if registration is an imperfect proxy for whether boards keep functioning. Consistent with this reading, the effect concentrates where civic representation was already dense in 1980, before the

modern conflict, so the targeting incentive scales with the density of the network rather than the individual registration event. Together these two findings map both the threat and its limits and point toward collective rather than individual protection as the relevant policy lever, bridging the literatures on armed governance (Arjona, 2016), civic organizations in conflict (Kaplan, 2017), and postaccord violence in Colombia (Albarracín et al., 2023a).

2 Related Literature

2.1 Civic organizations, collective action, and violence

The civic-capital tradition invoked in the introduction rests on a broad empirical base: Varshney (2003) shows that inter-ethnic civic engagement suppresses communal violence in India; Bellows and Miguel (2009) document that communities with stronger pre-war social networks recover collective action capacity faster after the Sierra Leone civil war; and Casey et al. (2012) show that community-driven development builds social capital and reduces conflict exposure in the same setting. The closest existing work on the Colombian setting is Kaplan (2017), who exploits a cross-sectional snapshot of junta density measured in 1985 across Colombian *municipios* and finds a robust negative association with general municipal homicide rates over 1990–2005. The present paper differs on three margins that together explain why our baseline estimates have opposite signs without contradicting each other. The outcome here is the strict subset of homicides corresponding to targeted killings of social leaders, on which the JAC framework can simultaneously reduce ambient predation and raise the salience of community leaders as a class, two effects that are not separately identifiable from a general-homicide outcome. The treatment is a time-varying flow of newly registered JACs identified within municipality from a 65-year administrative panel, rather than a 1985 stock compared between municipios. Kaplan’s own evidence is consistent with the targeting margin documented here: he hand-codes 546 killings of junta leaders from CINEP press records over 1996–2006 with a marked spike around 2000–2001, and reports a sharp attenuation of the protective association in his annual cross-sectional estimates from 2001 onwards. His design, with a treatment fixed in 1985 and an aggregate-homicide outcome, is not built to identify whether new JAC registrations causally generate this targeting; this is the question the present paper addresses. The contribution is to characterize the targeted-violence margin under within-municipality identification, not to overturn the protective finding on general homicides.

2.2 Armed governance, territorial control, and the targeting of social leaders

Kalyvas (2006) establishes that selective violence is a function of territorial control: groups target civilians selectively only when they control territory enough to tell collaborators from opponents. Arjona (2016) extends this to governance arrangements, distinguishing *rebelocracy*, in which the armed group intervenes broadly in civilian affairs, from *aliocracy*, in which it tolerates pre-existing civilian institutions and limits itself to security and taxation: groups with stable control develop an institutional interest in regulating the civic life of the communities they govern. Staniland (2014)

shows that armed groups with dense pre-existing social networks sustain the cohesion such governance requires, a micro-foundation for the durability Arjona theorizes. Durable presence does not, however, mechanically imply violence against civic leaders: co-optation is typically cheaper (Dal Bó et al., 2006; Gáfaró et al., 2022), and Gáfaró et al. (2022) document that armed groups co-opted JAC presidents as information intermediaries in some municipalities. Dal Bó et al. (2006) model the underlying co-optation-versus-elimination decision as a function of costs and rents; targeting becomes attractive precisely when a civic institution is state-recognized and externally accountable, and therefore hard to absorb, so the killings studied here are the observable residual of a strategic choice, not its default. On the Colombian case, Prem et al. (2022) show that partial peace agreements raise selective civilian targeting by disrupting established territorial arrangements; Albarracín et al. (2023a) document the pathways of post-accord violence, and Albarracín et al. (2023b) link the assassination of social leaders to local competitive authoritarianism; Arjona et al. (2025) establish a downstream cost, political assassinations reduce electoral turnout. This paper provides the first large-scale quantitative test of a prediction that follows from this framework: where governance horizons are long, registering a JAC is not merely an organizational event but a signal of rival authority, activating a targeting incentive absent in episodic or non-governance contexts. As in Ansell and Samuels (2014)’s analysis of incumbent elites under democratization, what governance-seeking armed actors resist is not individual leaders but the rise of organized rival institutions capable of contesting their authority.

2.3 Local institutions and armed conflict in Colombia

Several papers study the interaction between local institutions and armed presence in Colombia specifically. Gáfaró et al. (2014) show that the presence of armed groups reconfigures community collective action: organizations in zones of armed presence orient their activity toward the armed group rather than the state, documenting that institutional behavior responds endogenously to the governance structure of armed actors. Galindo-Silva (2021) shows that the opening of local political councils affects armed conflict intensity, establishing that institutional change at the community level generates strategic responses from armed actors. Dube and Vargas (2013) identify commodity price shocks as a driver of armed group activity, providing a benchmark for within-municipality variation in conflict exposure that motivates the identification design used here. This paper extends this literature from the question of how conflict shapes institutions to the reverse: how civic institutional expansion shapes the risk of targeted violence, exploiting a 65-year administrative panel that substantially extends the temporal scope available to existing work.

2.4 Contribution

Across all three strands, the contribution is the following: using within-municipality variation over 65 years, the paper shows that civic expansion can raise the risk of targeted killings of local leaders, that this risk is concentrated where armed governance is durable, and that targeting nonetheless fails to stop new boards from registering.

These findings reframe how we should think about civic expansion in contested territories. Civic expansion is not only a development intervention; in zones of armed governance it is also a move in a strategic contest over who organizes local social life, one that governance-seeking armed groups have strong incentives to resist.

The remainder of the paper proceeds as follows. Section 3 describes the institutional context and develops the conceptual framework. Section 4 presents the data and documents the determinants of civic expansion. Section 5 sets out the empirical strategy, reports the average total effect, and presents the robustness checks. Section 6 turns to heterogeneity by armed governance (first contribution) and to the institutional resilience of civic organization under targeting (second contribution). Section 7 concludes.

3 Institutional Background and Conceptual Framework

3.1 *Juntas de Acción Comunal*: History, Structure, and Functions

The *Juntas de Acción Comunal* were created by Law 19 of 1958 in the aftermath of *La Violencia*, the decade-long civil conflict between Liberal and Conservative factions, with an estimated number of fatalities of 164,270 between 1948 and 1958 (Oquist, 1978). The design logic was explicitly civic reconstruction: the law created a legal framework within which communities could self-organize into recognized neighborhood boards, channel collective energy toward development rather than political violence, and establish an institutionalized intermediary between dispersed rural communities and a state apparatus with limited reach. Communities chose whether and when to register under this framework; the state provided the enabling conditions, not the organizations themselves.

Law 743 of 2002 (Función Pública de Colombia, 2002) consolidated this framework, standardizing JAC statutes, election procedures, and oversight. Each JAC must hold elections for a president and governing board every four years, keep public accounts of the resources it receives and spends, and register with the Ministry of Interior. Every registration thus creates a JAC president: a named, publicly listed, legally constituted figure who signs official documents, appears in municipal records, and sits on municipal planning councils. This visibility is not incidental to the JAC framework but its defining feature, and it is the same registration requirement that generates the administrative data used in this paper.

In practice, JACs constitute the most granular layer of institutionalized civic life in Colombia. Each JAC covers a *vereda* (rural hamlet) or urban neighborhood. Their functions span community development (formulating and executing small public works), political mediation (transmitting community demands to municipal governments), and social organization (managing collective resources, resolving disputes). Figure 1 documents that the pre-determined JAC stock of 1980 is positively associated with literacy and GDP per capita, and negatively with unsatisfied basic needs and Gini inequality over 1995–2015, patterns that persist after controlling for population, consistent with a large literature on civic capital and development (Putnam, 2000). The goal of this paper is not to estimate the development effects of JACs, but to use the variation in their creation as a

window on the strategic dynamics that civic expansion generates in contested territories.

Several features distinguish JACs from other community-level institutions in Colombia and motivate their use as the unit of analysis. First, the JAC framework is universal and territorial rather than thematic: a JAC corresponds to a place (a *vereda* or neighborhood), not to a sector or productive activity. This contrasts with cooperatives, producers' associations, victims' associations, or indigenous and Afro-Colombian councils (*cabildos*, *consejos comunitarios*), which exist only where the relevant economic activity or ethnic community is present and cannot generate the within-municipality variation in civic infrastructure that this paper exploits. Second, JACs are the most granular and most numerous civic organization: the 48,767 JACs in the panel cover essentially every rural hamlet and most urban neighborhoods of the country, whereas no other registered organization type achieves comparable coverage. Third, the institutional visibility of the JAC president is the defining feature: every registration creates one publicly named figure who signs municipal documents and attends planning councils. This is precisely the observable margin through which civic expansion can become legible, and targetable, to armed actors, and it is the margin most directly captured in the social leader killings data. Fourth, the JAC framework has been legally stable and homogeneous since Law 19/1958, with the major refinements of Law 743/2002 operating within an unchanged institutional skeleton; this makes JAC registration counts comparable over a 65-year window in a way that no other Colombian civic institution allows. Finally, the registration data come from a single administrative source, the Ministry of Interior, with consistent recording practices across municipalities and over time, in contrast to the fragmented records that characterize most other organizational types in Colombia.

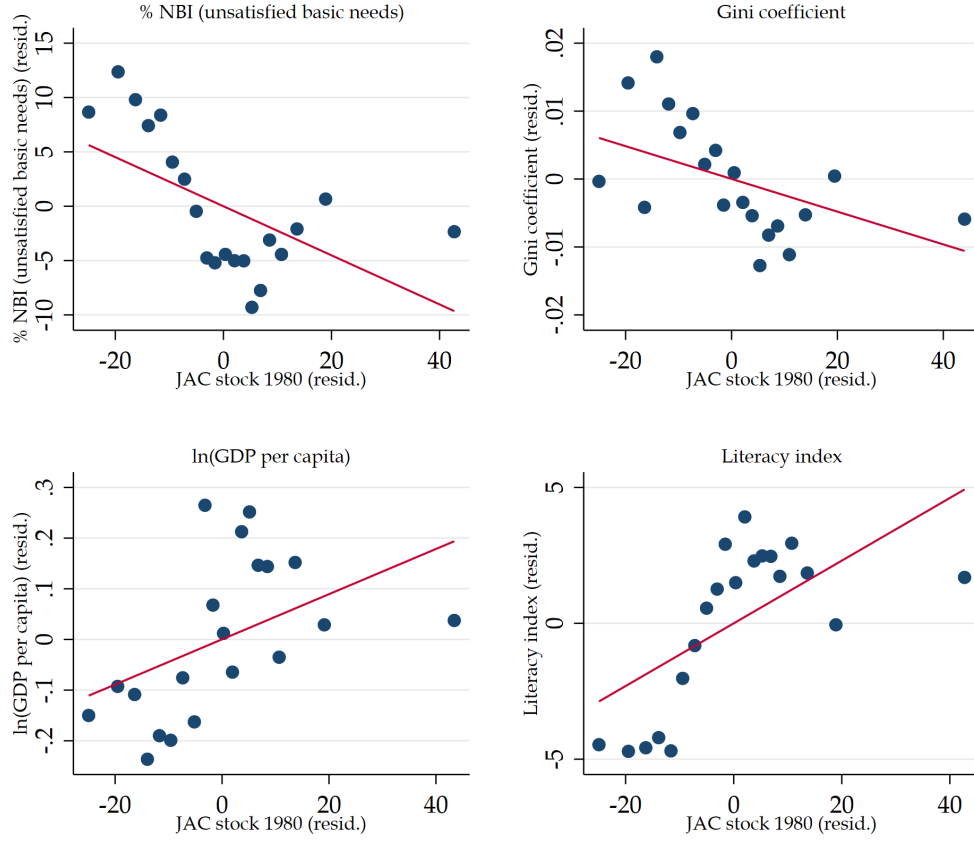


Figure 1: Pre-determined JAC stock (1980) and wellbeing indicators, net of population (binscatter, 1995–2015)

Notes: Partial binscatter (20 bins). Both outcome and JAC stock residualized on population (ln). Pre-determined stock (1980) predicting mean outcomes (1995–2015). Source: DANE, Ministry of Interior.

3.2 Armed Groups, Territorial Control, and Social Leader Killings

Colombia's internal armed conflict has set the state's armed forces against a succession of non-state actors. The state, through its military and police, has been a central party to the conflict throughout, but the targeting dynamic studied here runs through the non-state side, which has taken three principal forms: guerrillas (FARC, ELN), paramilitary organizations (AUC and successors), and their criminal successors (*grupos armados organizados*) (International Committee of the Red Cross, 2024). Each of these non-state actors has pursued territorial control as an organizational objective, not merely to extract rents from illicit crops or natural resources but to govern local populations: taxing economic activity, regulating social behavior, adjudicating disputes, and supplying rudimentary public goods (Arjona, 2016). This governance ambition distinguishes Colombia's armed groups from purely predatory criminal organizations.

The relationship between armed groups and JACs is empirically documented: Gáfaró et al.

(2022) show that armed groups co-opted JAC presidents in some municipalities as information intermediaries, while Arjona (2016) documents that groups with stable control sometimes delegated governance functions to community institutions. Dal Bó et al. (2006) model the underlying decision between co-optation and coercion as a function of the relative costs of each strategy, a logic that applies directly here: this paper tests the conditions under which targeting replaces accommodation as the dominant strategy. Because the data record only the killing margin, a co-opted president, alive but captured, is not observed as an event; the estimated effect therefore bounds institutional capture from below, missing the JACs neutralized through co-optation rather than violence.

Targeted killings of social leaders, recorded and classified by the CMH, are the observable outcome of this strategic dynamic. Of the 4,457 killings recorded by the CMH over 1960–2024, approximately 54 % are classified as *líderes comunitarios* (CMH code 50), the category that most directly corresponds to JAC presidents and board members (see Appendix B for the full distribution across CMH categories). This category is broader than JAC-related leaders: it also covers neighborhood organizers and communal volunteers who never held a board position, so the outcome is best read as a class-level proxy for community leadership rather than a JAC-office proxy. This is consistent with the paper’s reading that leaders are targeted for their role in the civic network, not as holders of a particular title. Figure 2 shows the annual count of targeted killings in Colombia since 1960, both for all social leaders (green line) and excluding community leaders (purple line). Both series track the same conflict cycles, suggesting that the temporal dynamics of targeted killings are driven by general armed conflict escalation rather than by factors specific to community leaders.

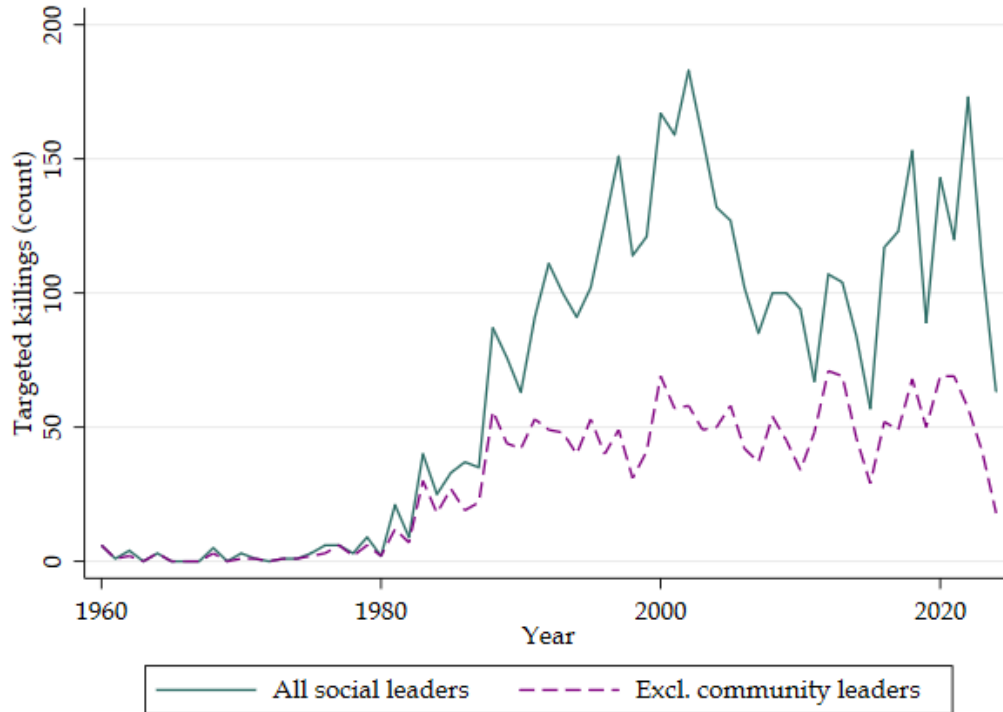


Figure 2: Targeted killings of social leaders per year

Notes: Annual count of targeted killings of social leaders (CMH, 1960–2024). Green line: all social leaders. Purple line: excluding *líderes comunitarios* (CMH code 50). Both series track the same conflict cycles. Source: Centro de Memoria Histórica.

3.3 Conceptual Framework

The starting point is Arjona (2016)’s distinction between armed groups that establish durable territorial governance and those engaged in episodic extraction. Groups that achieve stable rebelocracy regulate social behavior, adjudicate disputes, supply rudimentary public goods, and actively shape the institutional landscape of the communities they control. In doing so, they acquire a long-run stake in perpetuating their governance arrangement. Drawing on an analogy with Ansell and Samuels (2014)’s account of incumbent elites under autocracy, a governance-seeking armed group perceives a new JAC not primarily as a military threat but as an institutional one: each registration creates a named, legally constituted, publicly accountable body that organizes the community through state-recognized channels and signals that an alternative source of local authority exists. This institutional challenge activates a targeting incentive only for groups with sufficiently long governance horizons, those that internalize the full future cost of a rival civic institution. Groups with episodic or predatory presence have little strategic interest in eliminating institutional competitors; the costs of targeting visible leaders, retaliation risk, reputational damage, loss of community cooperation, outweigh any short-run benefit when the group cannot capture the long-run rents that stable governance provides (Kalyvas, 2006). It is thus not the presence of armed actors that acti-

vates the targeting mechanism, but the durability of their governance, operationalized empirically as three or more years of documented armed activity prior to 2016. Two qualifications follow. First, in zones of consolidated rebelocracy some JACs are created with the armed group’s acquiescence or under its instigation (Arjona, 2016; Gáfaró et al., 2022); such group-sanctioned registrations pose no institutional challenge and should generate no targeting, diluting the estimated effect toward zero. Second, co-optation is a substitute for elimination that the killing data do not record. Both forces imply that the estimates understate the strategic response of armed groups to civic expansion.

The framework decomposes the effect of JAC expansion into two channels operating in opposite directions. The exposure channel operates through institutional visibility: each new JAC registration creates identifiable figures whose elimination removes a node of rival civic authority. This channel is activated only under durable armed governance. The protection channel operates unconditionally: civic expansion strengthens community monitoring capacity and raises the social cost of violence (Putnam, 2000). The framework yields three testable predictions. First, the average effect of JAC expansion on targeted killings should be positive where armed governance is prevalent. Second, the effect should be absent or negative in peaceful municipalities where the protection channel operates without countervailing incentives. Third, the effect should be significantly stronger under established armed control than under sporadic presence. Two additional predictions, that the effect is larger for community leaders than for other social leaders, and that the targeting incentive increases with the pre-existing JAC stock, are tested in Section 5.4.3 and Section 6.4 respectively.

4 Data and JAC Expansion

4.1 Data Sources

The unit of observation is the municipality-year. The panel covers 1,124 Colombian municipalities over 1960–2024, yielding 73,017 observations. Municipality identifiers follow the DANE DIVIPOLA coding. The core data source is the Ministry of Interior’s administrative registration database, which records the legal founding date and geographic location (municipality) of every JAC in Colombia. I aggregate individual JAC records to the municipality-year level to construct the annual flow of new JAC registrations per 10,000 inhabitants, the main treatment variable, normalized by DANE population estimates.

The outcome variable is the annual count of targeted killings of social leaders, drawn from *Centro de Memoria Histórica* administrative records and aggregated to the municipality-year level. Because the outcome is highly sparse (96.1 % zeros), I use $\ln(1+SLK/10,000)$ as the main dependent variable, with robustness to the inverse hyperbolic sine transformation and raw levels.

Conflict presence is measured using two complementary indicators. The first is a time-invariant binary indicator equal to one if the municipality recorded at least one documented attack by FARC, ELN, or AUC at any point during 1993–2024, drawn from the CEDE Municipal Panel (Universidad

de los Andes); it classifies 845 municipalities (75 %) as having documented armed-group presence and is used for descriptive purposes. The second characterizes the nature of armed presence over 1993–2015, entirely predetermined relative to the post-2016 outcome window, measured as the number of years in which each municipality recorded at least one armed action, and classified as no presence (zero active years), sporadic (1–2 active years), or established (3+ active years), following Arjona (2016)’s distinction between extractive incursions and durable territorial governance.

Two measurement limitations bear noting. First, this classification is constructed from documented armed actions rather than from direct observation of governance arrangements; it approximates the persistence of armed presence rather than its governance character. Municipalities classified as established-presence zones maintained a durable armed foothold but did not necessarily achieve the full social contract that rebelocracy requires. Second, both the CEDE conflict data and the CMH outcome records may have imperfect coverage in remote municipalities, precisely the places most likely to host established armed-group presence and the highest true violence. This measurement error adds noise and, insofar as underreporting concentrates in these established-presence, high-violence municipalities, attenuates the gradient, biasing against the heterogeneity documented in Section 6.3. These channels, like the co-optation and group-sanctioned registrations discussed in Section 3, push the estimate toward zero. The one threat that is not signed in that direction, a time-varying local confound that could instead inflate the association, is probed, though not settled, by the permutation test of Appendix G. The headline estimate is therefore conservative with respect to measurement and substitution, rather than a lower bound in every dimension. Table 1 summarizes descriptive statistics of the data used in the present study.

The panel covers 1,124 municipalities observed annually from 1960 to 2024, or 73,017 municipality-years. Targeted killings are rare in level terms: 4,405 killings in total, an average of 0.060 per municipality-year, with no killing recorded in 96.1% of municipality-years, which motivates the $\ln(1 + y)$ outcome. They are widespread yet concentrated: 633 municipalities (56.3%) record at least one killing, while the 57 most affected account for 45.7% of the total and 69 municipalities concentrate half of it. Civic expansion is by contrast pervasive, with 48,767 new boards registered and 1,093 municipalities treated at some point.

Table 1: Descriptive statistics

Statistic	Value
<i>Panel A: Sample</i>	
Municipalities	1,124
Years	1960–2024
Municipality-years	73,017
Mean population per municipality	31,858
Median population per municipality	10,816
<i>Panel B: Targeted killings of social leaders (SLK)</i>	
Total killings (1960–2024)	4,405
Mean killings per municipality-year	0.060
Mean SLK per 10,000 inhabitants	0.025
P90 SLK per 10,000 inhabitants	0.000
Share of zeros	96.1%
Municipalities ever affected	633 (56.3%)
Share of killings in top 5% of municipalities	45.7%
Municipalities covering half of all killings	69
<i>Panel C: JAC creation (treatment)</i>	
Total JAC created (1960–2024)	48,767
Ever-treated municipalities	1,093
Share treated municipality-years	28.3%
<i>All municipality-years:</i>	
Mean new JAC (count)	0.684
SD new JAC (count)	1.735
P90 new JAC (count)	2.000
Mean new JAC per 10k	0.500
SD new JAC per 10k	1.491
P90 new JAC per 10k	1.542
<i>Treated municipality-years only:</i>	
Mean new JAC (count)	2.361
P90 new JAC (count)	5.000
Mean new JAC per 10k	1.695
P90 new JAC per 10k	3.919

Notes: Panel A: full sample. Panel B: outcome (social leader killings, all categories). Panel C: JAC treatment. Source: Centro de Memoria Histórica (killings); Ministry of Interior (JAC); DANE (population). The total of 4,405 killings in Panel B reflects the panel-level aggregate after collapsing individual records to the municipality-year level and restricting to the 1960–2024 window with valid municipality identifiers; the figure of 4,457 in the text reflects the raw record count from the CMH administrative database prior to panel construction, which includes a small number of observations with incomplete geographic coding.

4.2 The Treatment: JAC Creation and Its Distribution

The treatment variable, the annual flow of new JAC registrations per 10,000 inhabitants, exhibits two features that are important for the empirical strategy and for interpreting the heterogeneity

results. First, the variation is broad: 1,093 of Colombia’s 1,124 municipalities were eventually treated, with a mean flow of 1.70 new JACs per 10,000 among treated municipality-years and a P90 of 3.92. The within-municipality variation in annual flows, the source of identification in the dynamic DiD, is substantial.

Second, treatment intensity is not concentrated in peaceful municipalities. A striking feature of the data is that established-presence municipalities create *more* JACs than sporadic or peaceful ones: 28.1 % of municipality-years in established-presence zones are treated, compared to 18.7 % in sporadic zones and 13.6 % in no-presence zones. While treatment intensity is higher in conflict municipalities, all three groups exhibit meaningful within-municipality variation in JAC creation timing.

4.3 Timing of JAC Creation

Figure 3 plots annual JAC registrations at the national level over 1960–2024. Registration is uneven over time: a large early expansion in the 1960s–1970s, a second relative rise around the 1991 Constitution, and a clear spike localized after the 2016 peace agreement. The identifying assumption is not assessed from this aggregate pattern but from the determinants of JAC formation (Section 4.4) and the timing diagnostics (Section 5), which test directly whether JAC creation responds to conflict.

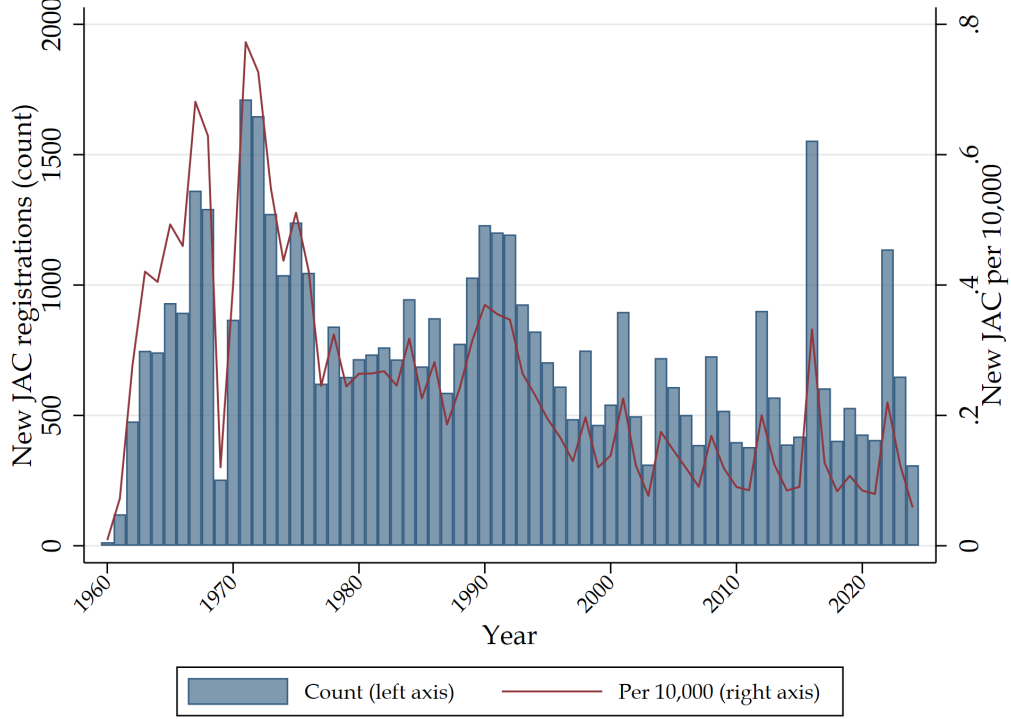


Figure 3: Annual JAC registrations: national aggregate 1960–2024

Notes: Bars: total new JAC registrations in Colombia per year (left axis). Line: new JAC registrations per 10,000 inhabitants at the national level (right axis).

4.4 What Drives JAC Creation?

To document the determinants of civic expansion and assess the plausibility of identification, I use cross-sectional post-double-selection LASSO regressions of JAC stock at five points in time (early stocks: 1970, 1975, 1980; late stocks: 2016, 2024) on a rich set of geographic, historical, and demographic controls, partialling out altitude, region dummies, log population, and distance to Bogotá. The predictor set includes geographic variables (area, distance to the departmental capital, distance to the nearest market, urban share, urban classification), pre-colonial and colonial variables (indigenous population density circa 1535, Spanish settlement intensity 1510–1561, municipal founding date), and historical conflict variables (*La Violencia* 1948–1953, land conflicts 1901–1917 and 1918–1931). For late stocks only, conflict variables capturing post-1993 armed presence are added to the penalized candidate set.

Table 2 reports the selected variables. For early stocks (1970–1980), the LASSO selects only distance to the departmental capital: municipalities further from administrative centers adopted the JAC framework earliest. Historical conflict variables, *La Violencia*, land conflicts 1901–1931, colonial settlement, are never selected for early stocks. By contrast, for late stocks (2016 and 2024), modern conflict variables enter the selected set despite penalization, documenting that the conflict has reshaped the geography of JAC density from 2016 onwards. This pattern is important for

identification, with one nuance. The LASSO exploits *between*-municipality variation in stocks: early JAC formation was driven by geography and state distance rather than by conflict history, consistent with a demand-driven diffusion process, while the late-stock results confirm the descriptive fact that more conflict-exposed municipalities have accumulated more JACs by 2016–2024, so cross-sectional stock levels are contaminated by the very conflict dynamics whose effect the paper seeks to identify. *Within* municipalities, by contrast, the dynamic event study of Section 6.5 shows that past killings do not affect the timing of JAC creation. The two facts are not contradictory: they operate at different levels of variation, and together they motivate within-municipality timing variation, rather than cross-sectional stock levels, as the source of identification.

Table 2: Determinants of JAC stock: early (1970–1980) vs. late (2016, 2024)

Predictor	Early stocks			Late stocks	
	1970	1975	1980	2016	2024
<i>Geographic & structural predictors</i>					
Dist. dept. capital	-0.002	-0.002	-0.002	—	—
<i>Conflict variables (penalised set, late stocks only)</i>					
Armed presence ever (FARC/ELN/AUC)	—	—	—	0.159	0.130
Conflict intensity 1993–2015 (0/1/2)	—	—	—	0.160	0.175
Share established-presence neighbors	—	—	—	0.335	0.285

Notes: Post-double-selection LASSO (IHS transformation). Partialling out (unpenalized): altitude, region dummies, log population, distance to Bogotá. Early stocks (1970–1980): conflict variables not included in candidate set. Late stocks (2016, 2024): conflict variables added to penalized set, entry indicates LASSO selects them despite penalization, documenting post-treatment contamination. — indicates variable not selected or not in candidate set.

5 Empirical Strategy and Baseline Results

5.1 Identification

Identification relies on within-municipality variation in the intensity of JAC creation over time. The identifying assumption is that, conditional on municipality and year fixed effects and department-specific non-parametric time trends, the *timing* of JAC creation within a municipality is uncorrelated with time-varying unobservables affecting targeted killings. Municipality fixed effects absorb all time-invariant municipal characteristics, including pre-existing JAC stock, geographic remoteness, and historical conflict exposure. Year fixed effects absorb national-level shocks common to all municipalities. Department-specific non-parametric trends absorb slowly-moving regional differences in both JAC adoption patterns and conflict trajectories. The residual variation, deviations in annual JAC registration timing within a municipality relative to its own departmental trend, identifies the effect. Because targeted killings rise only from the mid-1980s, most of this identifying co-movement is post-1990; the cross-sectional LASSO of Section 4.4 therefore speaks to the geography of early adoption, while the credibility of the post-1990 timing assumption rests on the anticipation, reverse-causality, and conflict-index diagnostics below rather than on the early-stock evidence.

Six pieces of evidence support this assumption. The LASSO evidence of Section 4.4 shows that early JAC formation (1970–1980) is driven by geography and state distance, not by historical conflict. The pre-treatment coefficients of the event study show no pre-trend in killings ahead of JAC creation (joint placebo $p = 0.338$), ruling out anticipation, and the dynamic event study of JAC creation around killings finds no reverse causality from past violence (Section 6.5). A regression on a broad conflict index yields joint $p = 0.359$ (Appendix C); for electoral cycles, presidential indicators show some predictive power for JAC creation, but the main DiD effect is fully robust to including cycle dummies as controls (point estimate ranges from 0.00310 to 0.00318 across specifications, Appendix D). An intra-department permutation test ($B = 2,000$ iterations) is directionally consistent with a genuine within-municipality effect rather than a departmental-level confound, though its aggregate p -value (0.137) sits at the margin of conventional significance, a dispersion that is partly mechanical: the per-capita treatment ratio has highly heterogeneous variance across municipalities of very different population sizes, which widens the permuted null and lowers the power of the aggregate statistic (Appendix G). Taken together these diagnostics are collectively inconsistent with the main identification threats, while acknowledging that no single test is individually conclusive. Section 5.4.1 provides a summary table of the six diagnostics.

5.2 Dynamic DiD: Average Total Effect

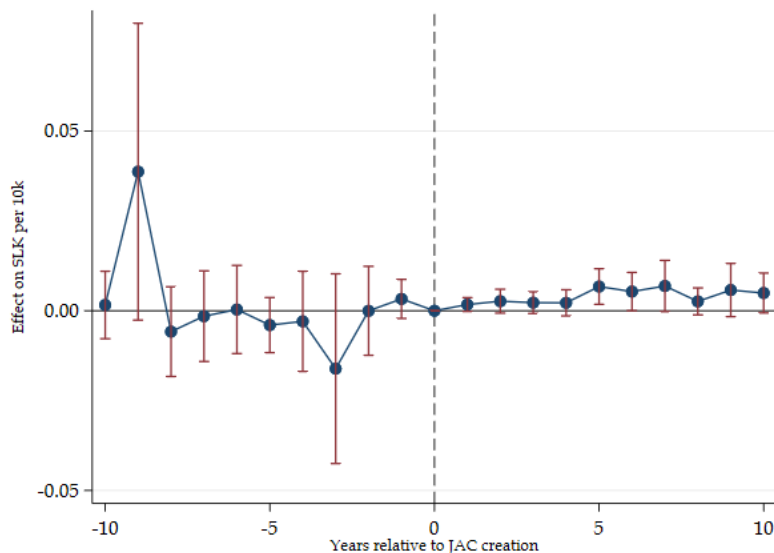


Figure 4: Dynamic DiD: average total effect of JAC creation on targeted killings

Notes: Dynamic event-study estimates with municipality and year fixed effects, department non-parametric time trends, and standard errors clustered at the municipality level. Treatment: new JAC per 10,000 inhabitants (no separate population control: the ratio normalizes mechanically for population scale).

To gauge economic magnitude, Appendix E evaluates the predicted effect at representative values of the JAC flow distribution. At the mean JAC flow among treated conflict municipality-years

(1.09 JACs per 10,000), the predicted increase in $\ln(1 + \text{SLK}/10,000)$ represents approximately 11.5% over the baseline mean in untreated conflict municipality-years; at the median (0.65 JACs per 10,000), the effect represents 6.8% of baseline; and at the seventy-fifth percentile (1.30 JACs per 10,000), 13.7%. As a benchmark, one additional new JAC per 10,000 inhabitants, the unit increase in the treatment variable, raises $\ln(1 + \text{SLK}/10,000)$ by 0.00314, corresponding to 10.5% of the baseline mean in conflict municipalities and 20.8% unconditionally. These magnitudes are non-trivial relative to a baseline outcome that is itself low: the violence is rare, and the marginal contribution of civic expansion to its incidence is correspondingly substantial. In levels, since $\ln(1 + y) \approx y$ for this sparse outcome, one additional JAC raises subsequent targeted killings by approximately 0.0035 per 10,000 inhabitants, about 0.011 additional killings in a municipality of average size, cumulated over the roughly six years across which the effect builds, or on the order of one additional killing for every ninety JACs created. Appendix E also evaluates the effect unconditionally across the full panel for comparison.

5.3 TWFE Comparison

For completeness, I also report a standard two-way fixed effects specification:

$$y_{mt} = \beta \text{JAC}_{mt} + \theta \ln(\text{Pop}_{mt}) + \alpha_m + \gamma_t + \delta_{d(m)} \cdot t + \varepsilon_{mt}, \quad (1)$$

where $y_{mt} = \ln(1 + \text{SLK}_{mt}/10,000)$; JAC_{mt} is the count of new JACs; Pop_{mt} is contemporaneous population; α_m are municipality fixed effects; γ_t are year fixed effects; $\delta_{d(m)} \cdot t$ are department-specific linear trends; and standard errors are clustered at the municipality level. Table 3 reports four specifications; the coefficient on JAC creation is positive and consistent in sign across all of them, but with full controls (municipality and year fixed effects and department-specific linear trends) it is small and imprecise. With continuous, staggered treatment, TWFE is subject to heterogeneity bias (de Chaisemartin and D’Haultfœuille, 2020): the negative weighting of heterogeneous effects leaves the sign of the bias undetermined, so this specification is an imprecise benchmark, not a causal estimate. The dynamic DiD of Section 5.2 is the primary estimate.

Table 3: JAC creation and targeted killings: TWFE estimates

	(1)	(2)	(3)	(4)
	(1) No FE	(2) Muni+Year FE	(3) +Dept trends	(4) Count+ln(pop)
Flow JAC (per 10k)	0.00016 (0.00044)	0.00062 (0.00039)	0.00060 (0.00042)	
Flow JAC (count)				0.00028 (0.00032)
Population (ln)				-0.00297 (0.00338)
Observations	68269	68269	68269	68269
Municipality FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
Dept $\times$ trend	No	No	Yes	Yes
Treatment unit	JAC/10k	JAC/10k	JAC/10k	Count + ln(pop)
Mean outcome (untreated)	0.0151	0.0151	0.0151	0.0151
SD outcome (untreated)	0.1022	0.1022	0.1022	0.1022

Notes: Outcome: $\ln(1 + \text{SLK per } 10,000)$, all social leaders. Treatment: count of new JAC registrations; contemporaneous population (ln) entered as a control. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Robustness

5.4.1 Summary of Threats to Identification

Before presenting the robustness checks, Table 4 consolidates the six diagnostics used to address the main threats to the identifying assumption. Each row maps a specific threat to its corresponding test and reports the baseline result, with cross-references to the underlying tables and discussion. The diagnostics fall into three groups. First, the LASSO evidence of Section 4.4 shows that early JAC formation (1970–1980) is driven by geography and state distance rather than by historical conflict (#1). Second, the event-study placebo coefficients rule out anticipation (placebo $p = 0.338$, #2), the resilience event study finds no effect of past killings on subsequent JAC creation (#3), and a joint F -test on three lags of a broad conflict index does not predict JAC creation ($p = 0.359$, #4); together these address reverse causality in both directions. The electoral-cycle confound (#5) is addressed differently: presidential indicators do predict JAC creation, but the main DiD estimate is essentially unchanged when cycle controls are included (0.00310 vs. 0.00314). Third, the intra-department permutation test reported in Appendix G (#6) addresses the hypothesis that the within-municipality association reflects a departmental-level confound (DiD $p = 0.137$, $B = 2,000$ iterations). No single diagnostic is individually conclusive, but the collective pattern is inconsistent with the main observable identification threats.

Table 4: Threats to identification: summary of diagnostics

#	Threat to identification	Diagnostic	Result	Reference
1	JAC formation driven by historical conflict	LASSO selection on early stocks (1970–1980); conflict variables not selected despite penalisation	Conflict vars not selected	Table 2
2	Anticipation: armed groups react to future JAC creation	Pre-treatment placebo coefficients of the event study	Placebo $p = 0.338$	Fig. 4
3	Reverse causality: past killings drive JAC creation	Dynamic event study of JAC creation around killings	No effect	Fig. 5
4	Broader conflict dynamics drive JAC formation	Joint F -test on 3 lags of broad conflict index on JAC	$p = 0.359$	Table C3
5	Electoral cycles accelerate JAC registrations	Main DiD re-estimated with municipal and presidential cycle controls	Stable (0.00310 vs. 0.00314)	Table D4
6	Departmental-level confound generates spurious within-municipality association	Intra-department permutation test, $B = 2,000$ iterations	DiD $p = 0.137$	App. G

Notes: Each row reports a threat to the identifying assumption together with the corresponding diagnostic and baseline result. Diagnostics #2–4 are joint tests on lead or lag coefficients in within-municipality specifications with full controls; #5 reports the stability of the main estimate to electoral-cycle controls. Diagnostic #6 reports the intra-department permutation p -value for the DiD specification ($B = 2000$ iterations); the full test is reported in Appendix G. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4.2 Alternative Outcome Transformations

Table 5 confirms that the dynamic DiD estimator yields positive and statistically significant average total effects across all four outcome transformations. The point estimates range from 0.00314 for $\ln(1 + \text{SLK}/10,000)$ to 0.00649 for the three-year cumulative outcome; all 95% confidence intervals exclude zero. Joint placebo p -values are consistently above 0.10 across all specifications, providing no evidence of pre-trends. The robustness of both magnitude and precision across functional forms is reassuring: the result does not hinge on the specific transformation used.

Table 5: Dynamic DiD robustness: average total effect by outcome

Outcome	Av. total eff.	95% CI	Placebo p -value
$\ln(1+\text{SLK}/10\text{k})$	0.00314	[0.00123, 0.00506]	0.338
$\text{asinh}(\text{SLK}/10\text{k})$	0.00404	[0.00162, 0.00645]	0.329
SLK/10k (levels)	0.00522	[0.00201, 0.00843]	0.266
$\ln(1+3\text{y cum SLK}/10\text{k})$	0.00649	[0.00229, 0.01068]	0.798

Notes: Each row re-estimates the dynamic difference-in-differences specification with the indicated outcome: row 1, $\ln(1 + \text{SLK}/10,000)$ (baseline); row 2, the inverse hyperbolic sine $\text{asinh}(\text{SLK}/10,000)$; row 3, SLK/10,000 in levels; row 4, $\ln(1 + \text{three-year cumulative SLK}/10,000)$, which cumulates killings over years t to $t + 2$. Treatment: new JAC registrations per 10,000 inhabitants. All specifications include municipality and year fixed effects, department non-parametric trends, and standard errors clustered at the municipality level.

5.4.3 Alternative Outcomes

Two further outcome variations test the specificity of the effect on the targeted-violence margin. The first re-estimates the main specification after excluding *líderes comunitarios* (CMH code 50), the category most directly corresponding to JAC presidents and board members, to test whether the effect operates beyond the mechanically-created subset of leaders. The second uses as outcome the count of homicides of non-vulnerable individuals (CMH classification), constructed from the same source database but excluding all social leader categories.

Non-community leaders. Excluding *líderes comunitarios* reduces the average total effect to $\hat{\beta} = 0.00163$ (SE = 0.00076); placebo $p = 0.536$, approximately half the magnitude of the all-leaders estimate, still positive and statistically significant (Appendix H, Figure H2). This is a conservative test: a positive effect on the remaining categories of social leaders, peasant, indigenous, union, and other, despite excluding the mechanically-created subset is consistent with civic expansion raising the organizational salience of local leaders as a class. The decomposition cannot, however, cleanly distinguish the institutional salience channel from a general conflict-escalation channel with available data.

Non-vulnerable homicides (placebo). If the JAC–violence relationship reflected a general conflict-intensification channel, we would expect a similar effect on homicides of non-vulnerable victims. The point estimate is small and negative $\hat{\beta} = -0.00066$ (SE = 0.02633); placebo $p = 0.049$, but it is very imprecisely estimated: the standard error (0.0263) is roughly twenty-five times the baseline specification standard error, so the confidence interval is wide enough to contain both zero and effects as large as the baseline (Appendix H, Table H7). This placebo is therefore suggestive rather than decisive; the imprecision most plausibly reflects that selective killings of non-vulnerable victims are themselves sparse. Two caveats follow. The pre-trend test rejects at the 5% level (placebo $p = 0.049$, against 0.338 in the main specification), so parallel trends is itself questionable in this specification; and because the point estimate is near zero, the reading does not turn on that diagnostic. I therefore read the declining pattern across outcomes, from JAC-affiliated leaders, to other social leaders, to ordinary homicide victims, as consistent with the targeted-violence interpretation, but I do not rest the rejection of a general conflict-intensification channel on this placebo alone.

5.4.4 Variation in Treatment Specification

Two further specifications express the treatment as a raw count rather than a per-capita flow, controlling for log population directly. Specifying the treatment as the count of new JAC registrations yields an average total effect of $\hat{\beta} = 0.00352$ (SE = 0.0011, 95% CI [0.0014, 0.0056], placebo $p = 0.342$); a binary version equal to one if the municipality records any new JAC registration in a given year delivers $\hat{\beta} = 0.0089$ (SE = 0.0025, 95% CI [0.0039, 0.0138], placebo $p = 0.262$). Both are significant at the 1% level, and both joint placebo tests support parallel trends. Because the outcome $\ln(1 + \text{SLK}/10,000)$ is built on a sparse variable (mean 0.025 killings per 10,000, 96% zeros), $\ln(1+y) \approx y$ over the relevant range, so these coefficients read directly as changes in the killing rate, with the level interpretation given in Section 5.2 (on the order of one additional killing for every ninety JACs created). Expressed against the mean killing rate among untreated municipality-years, these correspond to increases of approximately 23% (count) and 59% (extensive margin); because such percentages are sensitive to the scale of a $\ln(1+y)$ transformation, the level interpretation is the more transparent statement of magnitude.

As a units-invariant check on the $\ln(1+y)$ transformation (Chen and Roth, 2024), I also re-estimate the effect with a fixed-effects Poisson (PPML) model of the killing count (Appendix I). The contemporaneous coefficient is positive but imprecise; cumulating over a six-year window raises the point estimate roughly twelvefold (from 0.0015 to 0.018), consistent with the dynamic build-up the event study documents, though the count model is far less precise and the cumulative effect is not significant at conventional levels ($p = 0.36$). The loss of precision is mechanical: the Poisson estimator drops every municipality that never records a killing, so it is identified on a substantially smaller sample than the level-outcome specification, which widens the standard errors. The sign and dynamics agree with the baseline, while the precise units-robust evidence remains the level-outcome DiD of Table 5 (SLK per 10,000, $\hat{\beta} = 0.00522$, significant at the 1% level).

6 Civic Expansion and Violence under Armed Governance

The framework predicts that the average effect conceals fundamental heterogeneity: the exposure channel should dominate only where armed groups pursue stable territorial governance. This section tests this prediction in two steps: a within-municipality dynamic DiD that estimates the average effect separately by conflict category, and a between estimator that documents the gradient by governance durability.

6.1 Identification Strategy for Heterogeneity

The heterogeneity analysis of this section makes no independent causal claim: it documents descriptive patterns whose coherence with the framework is the object of interest. The balance evidence reported in Appendix J shows that conflict and non-conflict municipalities differ substantially on pre-determined characteristics, including pre-existing JAC stock, geographic remoteness, and historical conflict exposure. A simple split on ever-conflict status therefore does not support a clean

causal comparison: any difference in the DiD estimates across groups could partly reflect differential pre-existing trends or selection on geographic characteristics. The primary identification strategy addresses this in two ways. First, it restricts the outcome to the post-2016 period, the years following the FARC peace accord, when targeting of social leaders reached its highest recorded rates. Second, it measures armed group character using data from 1993–2015, entirely before the outcome period. This design ensures that the moderator is predetermined relative to both the outcome period and any post-accord reorganization of armed groups.

6.2 Established vs. Sporadic Armed Presence

Table 6 reports average total effects estimated separately for each of the three conflict categories, across three outcome windows. Panel A restricts the outcome to the strictly predetermined post-2016 window (2016–2024), the cleanest specification because the moderator is measured entirely before the outcome period; Panels B (1993–2015) and C (full 1993–2024) show that the gradient is robust across windows. This heterogeneity is read as descriptive and suggestive of the governance-durability mechanism rather than as an independent causal test.

Table 6: Average total effects by pre-2016 conflict character, across three outcome windows

Group	Av. total eff.	SE	95% CI	Placebo p
<i>Panel A: 2016–2024 (post-accord window: strictly predetermined)</i>				
No presence (0 active years 1993–2015)	-0.03728	0.05322	[-0.14159, 0.06702]	.
Sporadic (1–2 active years)	0.00305	0.01232	[-0.02110, 0.02720]	0.393
Established (3+ active years)	0.02196	0.02932	[-0.03550, 0.07941]	0.271
<i>Panel B: 1993–2015 (pre-accord window)</i>				
No presence (0 active years 1993–2015)	-0.01132	0.01886	[-0.04828, 0.02563]	0.455
Sporadic (1–2 active years)	0.00533	0.01169	[-0.01758, 0.02823]	0.592
Established (3+ active years)	0.02129	0.02108	[-0.02002, 0.06260]	0.484
<i>Panel C: 1993–2024 (full window)</i>				
No presence (0 active years 1993–2015)	-0.00907	0.01510	[-0.03867, 0.02053]	0.455
Sporadic (1–2 active years)	0.00380	0.00834	[-0.01255, 0.02015]	0.435
Established (3+ active years)	0.02344	0.01811	[-0.01206, 0.05894]	0.647

Notes: Panel A restricts the outcome to the strictly predetermined post-2016 window (2016–2024), in which the conflict moderator (measured 1993–2015) is entirely predetermined; Panel B uses the pre-accord window (1993–2015) and Panel C the full panel (1993–2024), showing that the no-presence < sporadic < established ordering is robust across windows. Missing placebo p -values in Panel A reflect a mechanical constraint of the 9-year post-2016 window: the estimator requires sufficient pre-treatment periods for each switcher to compute the joint placebo test, which is not available on a panel as short as 2016–2024. Missing entries thus indicate that the test cannot be computed, not that estimation failed. Panels B and C, covering 23 and 32 years respectively, are unaffected. Each row reports a separate dynamic DiD estimation with municipality and year fixed effects and department-specific non-parametric trends; treatment is new JAC per 10,000 inhabitants. Standard errors clustered at the municipality level.

Three features of these estimates are notable. First, the point estimates follow the gradient predicted by the framework: established-presence municipalities show the largest effect, sporadic municipalities an intermediate one, and no-presence municipalities the smallest, exactly consistent with the prediction that governance-oriented armed actors generate stronger targeting incentives.

The directional ordering, not the precision of any single group’s estimate, is the testable implication of the framework; the complementary between estimator of Section 6.3 tests the same gradient on cross-sectional variation, where the established–sporadic differential reaches conventional significance in the strictly predetermined post-2016 window. Second, the gradient is robust across all three windows reported in Panels A, B, and C: the same ordering, no-presence < sporadic < established, obtains whether the outcome is restricted to the strictly predetermined post-2016 window or extended over the full 1993–2024 panel, indicating that the heterogeneity is not an artifact of any particular sample selection. Third, the joint pattern rules out the alternative that the conflict-municipality effect is driven by general insecurity: if conflict were driving targeting, we would expect no gradient between sporadic and established groups, since both categories experienced armed presence and the only feature distinguishing them is the persistence, and hence governance character, of that presence.

6.3 Between Estimator: Differential Elasticity by Conflict Character

To sharpen identification of the heterogeneity result, I complement the within-municipality DiD with a between estimator that exploits cross-sectional variation in mean JAC creation and mean killing rates across municipalities. The two estimators serve distinct inferential roles. The within-municipality DiD exploits *temporal* variation: do years in which a municipality creates more JACs see more killings, relative to the municipality’s own departmental trend? It identifies an average causal effect under the local parallel trends assumption. The between estimator exploits *cross-sectional* variation: do municipalities that create more JACs on average also have higher average killing rates, and is this relationship stronger in established-presence zones? It identifies the heterogeneity gradient under the stronger assumption that cross-sectional differences in mean JAC creation are exogenous to differences in mean killing rates, conditional on department fixed effects. The two estimators are complementary rather than substitutes: the within estimator provides the causal benchmark, the between estimator a well-identified descriptive quantification of the conflict-type gradient.

The cross-sectional exogeneity assumption is unlikely to hold strictly in a causal sense, and the between estimator is presented as complementary evidence rather than as an alternative causal estimate. Two considerations motivate its inclusion. First, the department fixed effects absorb a large share of the geographic sorting documented in Appendix J. Second, the moderator Estab_m is measured over 1993–2015, predating the post-2016 outcome window of Panel A and limiting the scope for reverse causation in that specification. The estimator should therefore be read as documenting coherence across levels of aggregation.

For each window, I collapse the panel by municipality (and by sub-period for the full window), yielding mean outcomes and mean treatment per municipality, and estimate:

$$\begin{aligned} \bar{y}_{m,p} = & \alpha + \beta \overline{\text{JAC}}_{m,p} + \gamma_1 \text{Spor}_m + \gamma_2 \text{Estab}_m \\ & + \delta_1 (\overline{\text{JAC}}_{m,p} \times \text{Spor}_m) + \delta_2 (\overline{\text{JAC}}_{m,p} \times \text{Estab}_m) + \mu_p + \lambda_{d(m)} + \varepsilon_{m,p}, \end{aligned} \quad (2)$$

where $\bar{y}_{m,p}$ is mean $\ln(1 + \text{SLK}/10,000)$; $\overline{\text{JAC}}_{m,p}$ is the mean count of new JAC registrations; Spor_m and Estab_m are indicators for sporadic and established presence (no presence is the reference); μ_p is a period fixed effect (only in the full-window panel, where each municipality contributes two observations); $\lambda_{d(m)}$ is a department fixed effect; and $\ln(\text{pop})$ enters as a control. The framework predicts $\delta_2 > \delta_1 \geq 0$.

Table 7: Between estimator: JAC–killings elasticity by conflict group

	(1) (A) 2016-2024	(2) (B) Full 1993-2024
JAC/10k (no presence, ref)	-0.00014 (0.00033)	0.00009 (0.00033)
JAC/10k $\times$ Sporadic	-0.00498 (0.00333)	0.00049 (0.00366)
JAC/10k $\times$ Established	0.01076 (0.00682)	0.01014** (0.00497)
Diff (Established - Sporadic)	0.01575 (0.00721)	0.00965 (0.00618)
SE		
p-value	0.029	0.118
Obs (muni $\times$ period)	1096	2194
Dept FE	Yes	Yes
Period FE	No	Yes
Sample	Post-2016	1993-2024

Notes: Between estimator. Outcome: mean $\ln(1 + \text{SLK}/10,000)$ per municipality $\times$ period. Treatment: mean new JAC per 10,000 inhabitants per municipality $\times$ period. Reference category: no armed presence. Each column estimated on its own collapsed sample; no separate population control (treatment already per capita). Department and period fixed effects. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 reports the between estimator for two windows: Panel A restricts to the post-2016 outcome period (moderator strictly predetermined relative to the outcome window); Panel B uses the full 1993–2024 panel. The interaction terms measure each group’s JAC–killings slope relative to the no-presence reference, so the framework’s prediction $\delta_2 > \delta_1 \geq 0$ maps directly onto them. Two features stand out. First, in the strictly predetermined post-2016 window (Panel A) the established-presence slope is the largest and the only positive one ($\hat{\delta}_2 = 0.01076$), and the established–sporadic differential is positive and significant (0.01575, $p = 0.029$): the core prediction $\delta_2 > \delta_1$ holds. The weaker part of the prediction, $\delta_1 \geq 0$, does not: the sporadic interaction is itself slightly negative and imprecise ($\hat{\delta}_1 = -0.00498$), so the three groups do not line up in a strict monotone gradient in this window; what the data support is the established–sporadic contrast, not a clean three-step ordering. Second, the established slope remains the largest and significant over the full 1993–2024 panel (Panel B, $\hat{\delta}_2 = 0.01014$), where the sporadic slope sits just above no-presence and the established–sporadic differential keeps its sign but loses significance (0.00965, $p = 0.118$). The established–sporadic differential is therefore positive in both windows, and significant precisely where identification is cleanest (Panel A). It is robust to the threshold used to define established presence (Appendix K).

The individual interaction coefficients in the predetermined windows are mostly imprecise, but

the established–sporadic differential, which is the framework’s actual testable implication, reaches conventional significance in the post-2016 window (Panel A, $p = 0.029$) and retains its sign in the full panel (Panel B). The between estimator is intrinsically less powered than the within estimator because the cross-sectional sample size is at most one observation per municipality per period, and the prediction concerns the directional contrast between governance types, not the magnitude of any single coefficient. That this contrast is positive in both windows in which the moderator is cleanly predetermined, using only cross-sectional variation, provides descriptive coherence with the within-municipality DiD result of Section 6.2.

A sharper and better-powered version of the same test enters governance durability *continuously*, as the number of active conflict years over 1993–2015, rather than binning municipalities into three groups. Table 8 interacts this duration with mean JAC flow in the between estimator. The interaction is positive and significant ($\hat{\beta} = 0.00153$, $p < 0.05$): each additional year of documented armed presence raises the elasticity of social-leader killings to civic expansion. The pattern is stable when the interaction is split around the 2016 accord, with comparable coefficients in the strictly predetermined pre-2016 (0.00244, $p < 0.01$) and post-2016 (0.00221, $p < 0.05$) windows, so it does not hinge on the three-year threshold used to define established presence. Entered continuously, durability traces the monotone gradient the framework predicts; this is the heterogeneity result on which I place the most weight, with the categorical split above serving as its intuitive illustration and the established–sporadic contrast its sharpest single comparison.

Table 8: Between estimator with continuous conflict-duration moderator

	(1) Pooled	(2) Split pre/post 2016
JAC/10k (no presence, ref)	-0.00159 (0.00147)	-0.00888** (0.00435)
Active years 1993–2015	0.00351*** (0.00037)	0.00336*** (0.00037)
JAC/10k $\times$ Active years	0.00153** (0.00069)	
JAC/10k $\times$ Active years $\times$ Pre-2016		0.00244*** (0.00092)
JAC/10k $\times$ Active years $\times$ Post-2016		0.00221** (0.00110)
Obs (muni $\times$)	2194	2194
Dept FE	Yes	Yes
Period FE	Yes	Yes

Notes: Between estimator. Outcome: mean $\ln(1 + \text{SLK}/10,000)$ per municipality $\times$ period. Continuous moderator: number of active conflict years 1993–2015. Column 1 enters the JAC $\times$ duration interaction directly; column 2 splits it around the 2016 accord. Department and period fixed effects. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, the within-municipality DiD, the between estimator (including its continuous-duration form), and the visibility-channel test of Section 6.4 converge on the same qualitative conclusion: the elasticity of social leader killings to JAC expansion is elevated under durable armed

governance and concentrated in municipalities with dense pre-existing civic infrastructure. The causal status of the three pieces of evidence differs: the within-municipality DiD rests on the local parallel trends assumption and is the primary causal estimate; the between estimator and the visibility-channel test provide descriptive coherence across levels of aggregation under explicitly stronger assumptions.

6.4 Initial JAC Density and the Targeting Environment

A complementary test of the stock-dependence prediction uses the predetermined JAC stock measured in 1980, thirteen years before the earliest conflict data and thirty-six years before the post-2016 outcome window, as a moderator. The framework predicts that the targeting incentive is increasing in the pre-existing density of civic infrastructure: a marginal new JAC raises institutional competition more in communities where civic representation is already dense than in those with sparse civic infrastructure. The 1980 stock is strictly exogenous to the post-1993 conflict dynamics that generate the main outcome variation, as documented in Table 2: it is driven by geography and state distance, not by historical conflict.

I split municipalities into two groups based on an indicator for the top tercile of JAC stock per 10,000 in 1980, and re-estimate the dynamic DiD separately on each group, in two windows. Panel A of Table 9 reports the full panel 1980–2024, restricted to the period over which the moderator is strictly predetermined for all observations. Panel B restricts the sample to conflict municipalities over 1993–2024.

Table 9: Average total effect by initial JAC density

Group	Av. total eff.	SE	95% CI	Placebo p
<i>Panel A: Full panel 1980–2024 (moderator predetermined)</i>				
Low JAC density (below top tercile, 1980)	-0.00374	0.00848	[-0.02037, 0.01289]	0.832
High JAC density (top tercile, 1980)	-0.00258	0.01675	[-0.03541, 0.03024]	0.986
<i>Panel B: Conflict municipalities only, 1993–2024</i>				
Low JAC density (below top tercile, 1980)	-0.00503	0.02736	[-0.05866, 0.04859]	0.415
High JAC density (top tercile, 1980)	0.05295***	0.01715	[0.01935, 0.08656]	0.894

Notes: Average total effect of JAC expansion by pre-1980 JAC density (top tercile vs. rest), a strictly predetermined moderator. Panel A: all municipalities, full panel; Panel B: conflict municipalities only. Treatment: new JAC per 10,000 inhabitants. Municipality and year fixed effects, department-specific non-parametric trends. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The contrast across panels is the test. In the full panel, the effect is small and statistically indistinguishable from zero in both density groups: there is no density gradient absent the targeting environment. Once the sample is restricted to conflict municipalities, the effect concentrates in high-density zones: it is significant at the 1% level for the top-tercile group and indistinguishable from zero in the low-density group, with placebos supporting parallel trends in both. Read alongside the between estimator of Section 6.3, this provides an independent test of the stock-dependence mechanism using a strictly predetermined moderator that is immune to the contamination concerns

raised by the post-2016 administrative wave.

6.5 Resilience: Targeting Does Not Suppress JAC Registration

A natural concern is that killings deter future JAC creation: communities exposed to violence might stop forming new boards, which would both complicate the interpretation of the main effect and imply that targeting succeeds in suppressing civic organization. I test this in two complementary ways, which point the same way.

First, regressing new JAC creation on lagged killings, the joint test that past killings predict subsequent JAC creation never rejects, at three- or five-year horizons, and whether killings are measured for all social leaders or for community leaders specifically (Appendix F, Table F6). Second, and method-consistent with the baseline, I estimate a dynamic event study with the de Chaisemartin et al. (2025) estimator, treating the onset of a killing as a non-absorbing binary event and new JAC creation as the outcome. Figure 5 shows the result: JAC creation does not fall in the years following a killing. The post-event coefficients are jointly indistinguishable from zero, $p = 0.81$ for all social leaders and 0.30 for community leaders, and the same conclusion holds under a two-way fixed-effects regression of JAC creation on lagged killings (Appendix F, Table F6). If anything, the community-leader point estimates are positive, the opposite of suppression.²

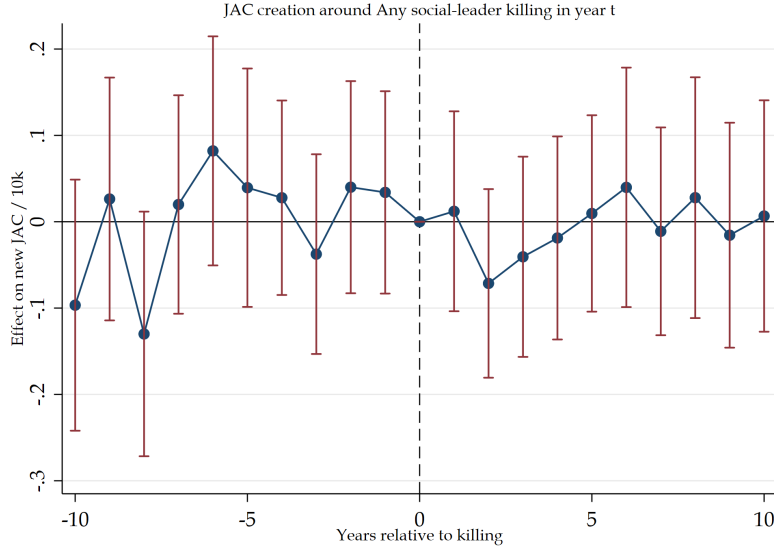


Figure 5: JAC creation around social-leader killings

Notes: Dynamic event study (de Chaisemartin–D’Haultfoeuille) of new JAC registrations per 10,000 inhabitants around the onset of a social-leader killing (binary, non-absorbing treatment). Municipality and year fixed effects, department non-parametric trends, standard errors clustered at the municipality level; 95% confidence intervals. Post-event coefficients are jointly indistinguishable from zero ($p = 0.81$); pre-event placebos show no pre-trend ($p = 0.28$).

²The community-leader specification rests on far fewer switching events than the all-leaders one and its pre-trend test is marginal (placebo $p = 0.093$); the all-leaders specification is cleaner (placebo $p = 0.284$). Both agree on the absence of a post-event decline.

JAC creation does not fall after targeting; new boards keep registering. This evidence is descriptive (killings are sparse and endogenous), but it indicates that targeted violence does not suppress the registration of new JACs. Registration is, however, a dubious proxy for civic functioning. A board can stay on the registry while its meetings lapse, its agreements with the municipality stall, and its leaders are replaced or co-opted, so this result bounds one margin, the persistence of registration, rather than showing that the civic network survives intact. Read this way, it also resolves an apparent puzzle: if eliminating a leader does not remove the institution, why do governance-seeking groups keep targeting? They may be suppressing a JAC’s effective functioning and its cooperation with the state while the bureaucratic act of registration continues. What invites targeting is a leader’s role in the network, not the individual.

7 Conclusion

This paper studies how the expansion of grassroots civic institutions shapes the risk of targeted violence against local leaders, using Colombia’s *Juntas de Acción Comunal* as a setting. Two findings emerge.

The central finding is that civic expansion increases targeted violence: at the mean treatment intensity in conflict municipalities, new JAC registration raises targeted killings of social leaders by about 11.5% over the baseline mean, roughly 0.13 additional killings per treated conflict municipality cumulated over the six-year horizon across which the effect builds, or on the order of 110 additional killings nationally across the 845 conflict municipalities in the sample, a result robust across outcome transformations and to excluding the post-2016 administratively contaminated zones. This average conceals marked heterogeneity: where armed presence is episodic or absent, the same expansion is protective or inconsequential, while the JAC–violence elasticity is highest where armed governance is durable, suggestive support for the governance-horizon mechanism, identified in the between estimator under assumptions stronger than the baseline DiD. Civic expansion is thus neither inherently protective nor inherently threatening; its effect is set by the governance character of the armed landscape in which it occurs.

The second finding bounds the threat. Targeted elimination of civic leaders does not suppress the registration of new JACs: creation does not fall after a social leader killing. Consistent with a stock-dependence mechanism, the elasticity concentrates in municipalities whose civic infrastructure was already dense in 1980, before the modern conflict period, indicating that targeting is invited by the density of the institutional network, not by the individual registration event; a placebo on non-vulnerable homicides is consistent with the incentive being specific to the targeted-violence margin. This margin, registration, is an insensitive proxy for whether a JAC keeps functioning, so the finding shows that targeting does not stop new boards from forming, not that the network is left intact. That distinction also dissolves an apparent paradox: governance-seeking groups may persist in targeting because they suppress a JAC’s effective functioning and its cooperation with the state even as registration continues. What motivates the targeting is a leader’s role in the

network, not the individual (Arjona, 2016).

The implication extends beyond Colombia. The civic-capital literature frames grassroots organizations as buffers against violence (Putnam, 2000); the evidence here shows that this buffer reverses where armed groups pursue territorial governance and read institutionalized civic representation as a rival claim to authority. As in Ansell and Samuels (2014)’s account of incumbent elites, what governance-seeking armed actors resist is not redistribution but the rise of organized institutions capable of contesting them.

The policy implication follows from the second finding, and it is not that civic expansion should be slowed. JACs deliver real local value, in development, public-goods provision, and the wellbeing gradients of Figure 1, and communities keep forming them even under fire; curtailing JAC registration or diluting the board’s mandate would forfeit those gains without removing the threat. Because the threat operates through organizational salience, the response is to protect leaders, not to retract the institution, and to do so at both levels at once. Individual protection, bodyguards and relocation, remains necessary but is insufficient on its own: targeting is invited by a leader’s role in the network rather than by their identity, so shielding one office-holder leaves the institutional pull that draws violence in place. The complementary and still underused lever is collective protection of the framework itself: legal recognition, security guarantees that protect board members as a class rather than one by one, and early-warning systems that treat the JAC network as the unit of risk. The stakes, quantified above, are real but bounded. Whether the 2016 accord itself caused the shift in targeting documented in Section 6 remains open: isolating FARC demobilization exposure from concurrent changes would require a dedicated design, and is a natural next step for research on peace agreements and targeted violence. A further avenue is disaggregation: the JAC is a *vereda*-level institution, and registration records carry sub-municipal information that the present municipal panel aggregates away. Matching killings to the specific community in which a board was created would tighten the link between an individual registration and the targeting it invites, and would allow the stock-dependence mechanism to be tested at the scale at which armed groups actually govern.

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A Robustness: Excluding PDET Municipalities

Table A1 reports the main DiD specification after excluding the 170 municipalities designated as priority PDET zones under Decreto 893 de 2017, the areas most likely to experience an administrative wave of JAC registrations driven by post-accord rural development programs rather than autonomous civic dynamics. The average total effect remains positive and precisely estimated (0.00205***, SE = 0.00079, placebo $p = 0.235$), attenuated relative to the full-sample estimate but qualitatively unchanged. This confirms that the main result is not driven by the post-2016 administrative contamination of the treatment variable documented in Table 2.

Table A1: Robustness: excluding PDET municipalities

Sample	Av. total eff.	SE	95% CI	Placebo p
Full sample (main estimate)	0.00314	0.00098	[0.00123, 0.00506]	0.338
Excl. PDET municipalities ($N = 954$)	0.00205	0.00079	[0.00050, 0.00359]	0.235

Notes: Each row re-estimates `did_multiplengt_dyn` on the full panel 1960–2024 with new JAC per 10,000 inhabitants as treatment (no population control). PDET municipalities: the 170 priority zones designated under Decreto 893 de 2017, which received an administrative wave of JAC registrations post-2016 as part of rural development programmes in former FARC territories. Excluding these municipalities tests whether the main result is driven by post-accord administrative contamination of the treatment variable. All specifications include municipality and year fixed effects and department non-parametric trends. SE clustered at municipality level.

B Distribution of Social Leader Killings by CMH Category

Table B2 reports the distribution of targeted killings across CMH leader categories over the full panel (1960–2024). *Líderes comunitarios* (code 50) account for approximately 54 % of all recorded killings and correspond most directly to JAC presidents and board members. The non-community leader outcome used in Section 5.4.3 excludes this category.

Table B2: Distribution of social leader killings by CMH category (1960–2024)

Leader category	N killings	Share (%)
Líder Comunitario (CMH code 50)	2421	54.3
Líder Campesino	480	10.8
Líder Indígena	454	10.2
Líder Sindical	380	8.5
Autoridad Indígena	242	5.4
Defensor de DDHH	214	4.8
Líder Cívico	127	2.8
Representante/Líder de Víctimas	86	1.9
Líder Estudiantil	30	0.7
Autoridad de Comunidades Negras	15	0.3
Líder Ambiental	8	0.2
Total (social leaders)	4457	100.0

Notes: Distribution of targeted killings of social leaders across CMH leader categories over the full panel (1960–2024). *Líderes comunitarios* (code 50) correspond most directly to JAC presidents and board members. Source: Centro de Memoria Histórica.

C Lagged Conflict Does Not Jointly Predict JAC Creation

Table C3 reports a regression of JAC creation on a broad conflict index, a binary indicator equal to one if the municipality records at least one documented armed action by FARC, ELN, or AUC in a given year, across the five event types recorded in the CEDE Municipal Panel, entered at lags 1 through 3. Only the third lag is individually significant, and only at the 10% level; the joint F -test yields $p = 0.359$, confirming that the within-municipality timing of JAC creation is not jointly predicted by prior conflict dynamics. To the extent that JAC creation responds to past conflict at all, that response is negative (the single marginally significant lag is negative). Combined with the positive association between conflict and killings, such a residual response would push the main estimate toward zero rather than inflate it. This complements the reverse-causality evidence of Section 6.5 (Table F6), which finds that past killings do not predict subsequent JAC creation.

Table C3: Lagged conflict does not jointly predict JAC creation

	(1) New JAC (per 10k)
Conflict index, year $t - 1$	-0.0008 (0.0154)
Conflict index, year $t - 2$	0.0055 (0.0167)
Conflict index, year $t - 3$	-0.0273* (0.0154)
Observations	31663
Joint p -value	0.359

Notes: Outcome: new JAC registrations per 10,000 inhabitants. Conflict index: binary indicator equal to one in years with at least one documented armed action (attacks on populations, political homicides, confrontations, armed contacts, kidnappings) by FARC, ELN, or AUC, entered at lags 1–3. All specifications include municipality and year fixed effects and department-specific linear trends. Standard errors clustered at the municipality level. Joint F -test p -value: 0.359.

D Electoral Cycles and the Robustness of the Main Effect

A potential identification threat is that years in which a municipality receives increased attention from the Ministry of Interior, due to a political clientelism cycle or a local election campaign, could simultaneously accelerate JAC registrations (through faster bureaucratic processing) and affect social leader killings (through disruption of local power equilibria).

I assess this threat in two steps, reported in the two panels of Table D4.

Panel A regresses JAC creation on municipal and presidential election-year indicators, inferred from administrative records that report voter turnout only in election years. Municipal election-year indicators are not significantly associated with JAC creation; presidential election-year indicators show a positive and significant association. The variation surviving year fixed effects comes from cross-sectional differences in turnout data coverage: certain municipalities have turnout records in some presidential election years but not others, possibly reflecting differential state administrative presence rather than a clientelistic cycle.

Panel B addresses the question that matters for identification: not whether electoral cycles correlate with JAC creation in absolute terms, but whether they confound the estimated relationship between JAC creation and social leader killings. It re-estimates the main dynamic DiD specification with three sets of cycle controls: a baseline without controls, municipal-cycle controls (last year, election year, and first year of mandate dummies), and presidential-cycle controls (analogous dummies). The dynamic DiD point estimate ranges from 0.00310 to 0.00318 across all three specifications, a variation of less than 3%, and the placebo joint p -values remain in the 0.32–0.36 range, providing no evidence of pre-trends. The within-municipality timing of JAC creation that drives the main result is therefore not contaminated by electoral cycle dynamics. Combined with the conflict-index test of Appendix C and the timing diagnostics of Appendix F, this constitutes six pieces of evidence that the within-municipality timing of JAC creation is uncorrelated with the

main observable threats to identification.

Table D4: Electoral cycles and the robustness of the main effect

<i>Panel A. Do election years predict JAC creation? (outcome: new JAC per 10k)</i>			
Election-year indicator	Coeff.	SE	<i>p</i>
Municipal election year	-0.00040	0.06397	0.995
Presidential election year	0.17085	0.08029	0.034
<i>Panel B. Dynamic DiD effect with electoral-cycle controls (outcome: $\ln(1 + \text{SLK}/10,000)$)</i>			
Cycle controls	Coeff.	SE	Placebo <i>p</i>
None (baseline)	0.00314	0.00098	0.338
Municipal cycle	0.00318	0.00097	0.324
Presidential cycle	0.00310	0.00093	0.358

Notes: *Panel A:* outcome is new JAC registrations per 10,000 inhabitants; election-year indicators inferred from non-missing turnout data; variation surviving year fixed effects comes from cross-sectional differences in turnout coverage. *Panel B:* outcome is $\ln(1 + \text{SLK}/10,000)$, treatment is new JAC per 10,000 inhabitants; the dynamic difference-in-differences specification uses ten effects and ten placebos with department non-parametric trends and standard errors clustered at the municipality level; cycle controls are last year of mandate, election year, and first year of mandate dummies (three dummies per cycle). All specifications in both panels include municipality and year fixed effects and department-specific linear trends. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Economic Magnitude: Predicted Effects at Representative JAC Flow Values

Table E5 evaluates the predicted effect of JAC creation at representative values of the JAC flow distribution. Panel A evaluates the effect conditional on conflict municipalities over 1993–2024, where the substantive mechanism is most active. Panel B evaluates the effect at unconditional moments of the JAC flow distribution across all treated municipality-years over the full panel 1960–2024, consistent with the estimation sample of the main dynamic DiD specification.

Table E5: Economic magnitude: predicted effects at representative JAC flow values

New JAC per 10k inhab.	$\Delta \ln(1 + \text{SLK}/10\text{k})$	As % of baseline	Baseline mean
<i>Headline effect: one additional new JAC per 10,000 inhabitants</i>			
1 JAC/10k (conflict baseline)	0.00314	10.53%	0.02985
1 JAC/10k (unconditional baseline)	0.00314	20.79%	0.01512
<i>Panel A: Conflict municipalities, 1993–2024, JAC/10k > 0</i>			
P25 (0.287 JAC/10k)	0.00090	3.0%	0.02985
P50 (median) (0.649 JAC/10k)	0.00204	6.8%	0.02985
P75 (1.304 JAC/10k)	0.00410	13.7%	0.02985
Mean (1.092 JAC/10k)	0.00343	11.5%	0.02985
<i>Panel B: All municipalities, full panel 1960–2024, JAC/10k > 0 (unconditional)</i>			
P50 (median) (0.974 JAC/10k)	0.00306	20.3%	0.01512
Mean (1.695 JAC/10k)	0.00532	35.2%	0.01512
<i>Notes:</i> Predicted effect = $\hat{\beta}_{\text{DiD}} \times (\text{new JAC per 10k})$, where $\hat{\beta}_{\text{DiD}} = 0.00314$ is the average total effect from the main dynamic DiD specification (treatment: count of new JAC registrations per 10,000 inhabitants; no population control as the ratio normalises mechanically) estimated on the full panel 1960–2024. Headline: one additional new JAC per 10,000 inhabitants raises $\ln(1 + \text{SLK}/10,000)$ by 0.00314, corresponding to approximately 10.53% of the baseline mean among conflict municipalities and 20.79% unconditionally. Panel A evaluates the effect at representative values of the JAC/10k distribution among treated conflict municipality-years (1993–2024); baseline mean is $\ln(1 + \text{SLK}/10,000)$ in untreated conflict municipality-years over the same window. Panel B evaluates the effect at unconditional moments of the JAC/10k distribution across all treated municipality-years (1960–2024), consistent with the estimation sample; baseline mean is $\ln(1 + \text{SLK}/10,000)$ in all untreated municipality-years over the full panel.			

F Reverse Causality: Past Killings and JAC Creation

Anticipation is addressed in the main text by the placebo coefficients of the main event study (joint placebo $p = 0.338$, Figure 4). This appendix collects the complementary evidence on reverse causality, the concern that past killings drive subsequent JAC creation. Table F6 regresses JAC creation on past killings using two estimators, a dynamic event study around killings and a two-way fixed-effects lag specification; neither detects an effect, and this is the basis of the resilience result of Section 6.5.

Table F6: Targeting does not suppress civic organization: two estimators

Killing event (treatment)	Dynamic event study (de Chaisemartin–D’Haultfoeulle)				TWFE lags
	Avg. effect	95% CI	Joint p	Placebo p	Joint p (3y/5y)
Any social-leader killing	-0.031	[-0.462, 0.399]	0.806	0.284	0.685 / 0.106
Any community-leader killing	0.328	[-0.352, 1.008]	0.303	0.093	0.557 / 0.577

Notes: Outcome: new JAC registrations per 10,000 inhabitants. Treatment: binary indicator equal to one in years with ≥ 1 killing (non-absorbing). Event-study columns from `did_multiplet_dyn`, municipality and year fixed effects, department non-parametric trends, ten effects and ten placebos, standard errors clustered at the municipality level; “Joint p ” tests joint nullity of the post-event effects, “Placebo p ” the pre-event coefficients (parallel trends / no anticipation). “TWFE lags” reports the joint F -test that three (resp. five) lags of killings predict JAC creation. Neither estimator detects an effect of past killings on subsequent JAC creation.

G Permutation Test: Intra-Department Treatment Reassignment

A residual identification concern is that a time-varying municipal-level shock, such as a local economic boom or a population influx, could simultaneously generate demand for JAC formalization and attract armed-group activity, producing a spurious positive correlation between JAC creation and targeted killings of social leaders even after conditioning on municipality fixed effects, year fixed effects, and department-specific non-parametric trends. The timing diagnostics of Section 5 address the observable component of this threat. This appendix addresses the unobservable component through a permutation test in the spirit of Conley and Taber (2011) and Young (2019).

In each of $B = 2,000$ iterations, the treatment, new JAC registrations per 10,000 inhabitants, is randomly reassigned across municipalities *within the same department*, preserving the departmental structure, including department-specific trends and composition, while breaking the within-municipality temporal correlation between JAC creation and targeted killings.³ Under the null hypothesis that the observed association reflects a departmental-level confound rather than a genuine within-municipality relationship, randomly reassigning treatment series within departments should produce coefficients of similar magnitude to the observed estimate. The permutation p -value is the share of permuted coefficients that exceed the real estimate in absolute value. Departments with fewer than ten municipalities are excluded to ensure that within-department redistribution is non-degenerate.

Figure G1 reports the results, and two features matter. First, the real event-study coefficients are positive at all ten post-treatment horizons and lie systematically above the placebo median, which is well-centered on zero (right panel), the directional signature of a genuine within-municipality effect rather than a departmental artifact. Second, the aggregate average total effect of 0.00314 falls in the upper tail of the permuted null but does not clear the conventional threshold (finite-sample $p = 0.137$, left panel). The permuted null is diffuse because the per-capita treatment ratio has highly heterogeneous variance across municipalities of very different population sizes, which mechanically widens the reassigned distribution and lowers the power of the aggregate statistic. Read against the consistently positive event-study coefficients, the permutation is directionally supportive of the main estimate while remaining, on the single aggregate statistic, at the margin of conventional significance. It is one of six diagnostics, each addressing a distinct threat and none individually decisive: the identifying inference rests on the clustered dynamic DiD, with the permutation serving as a conservative check against a departmental-level confound rather than as the primary test of the effect.

³Reassigning treatment series within departments preserves the departmental structure that the identification strategy conditions on, including department-specific non-parametric trends and the composition of municipalities within each department. What the permutation breaks is the within-municipality temporal correlation between JAC creation timing and outcomes. The test therefore directly addresses the concern that a departmental-level confound drives the observed association, while leaving intact the identifying variation that the main specification exploits.

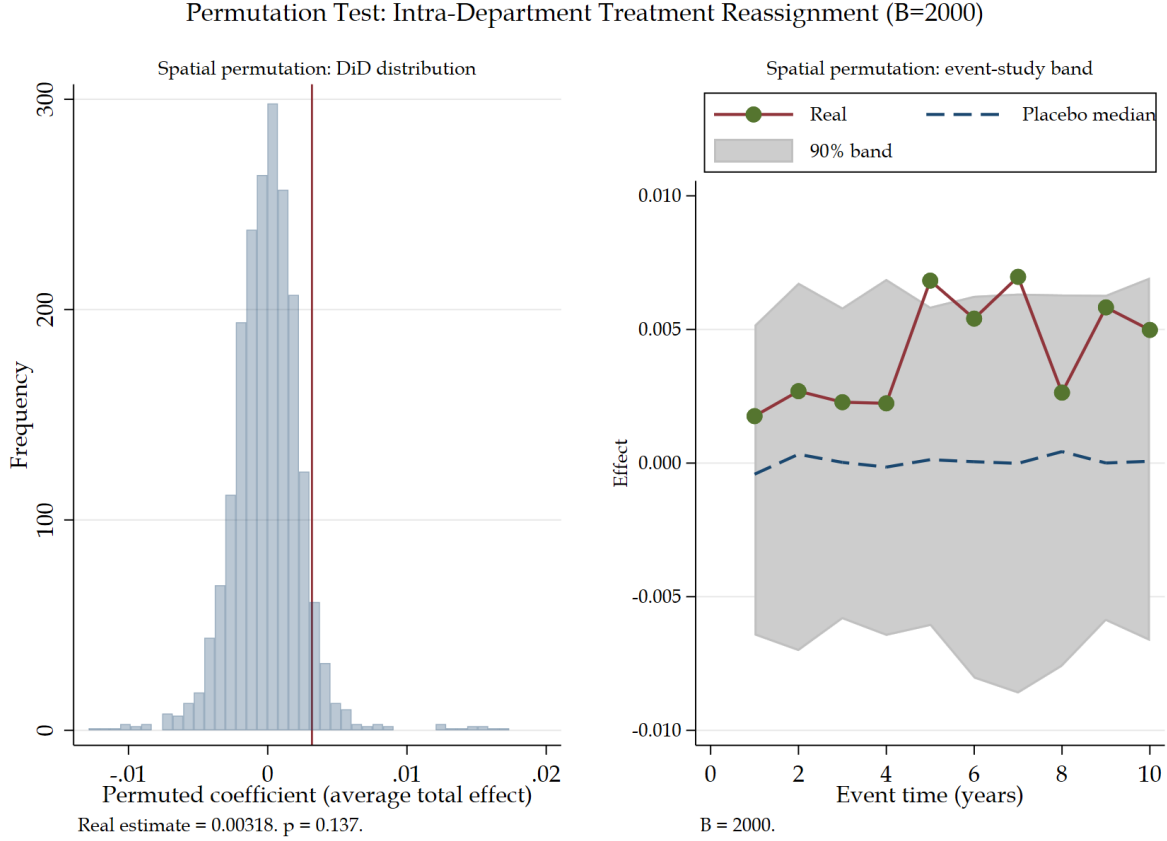


Figure G1: Permutation test: distribution of permuted coefficients

Notes: Left panel: distribution of DiD average total effects across $B = 2000$ intra-department permutations of the treatment (new JAC per 10,000 inhabitants); vertical line marks the real estimate (0.00314, permutation $p = 0.137$). Right panel: event-study placebo band (90% across permutations) versus real event-study coefficients (red); the placebo median (dashed line) is well-centered around zero. Departments with fewer than 10 municipalities excluded.

H Alternative-Outcome Event Studies

This appendix collects the exhibits behind the two alternative-outcome checks discussed in Section 5.4.3. Figure H2 re-estimates the main dynamic DiD after excluding *líderes comunitarios* (CMH code 50): the average total effect is about half the all-leaders estimate but remains positive, with no pre-trend. Table H7 reports the placebo specification with the outcome replaced by homicides of non-vulnerable individuals; the estimate is small, negative, and very imprecise, and the placebo pre-trend rejects at the 5% level, so this check is read as suggestive rather than decisive.

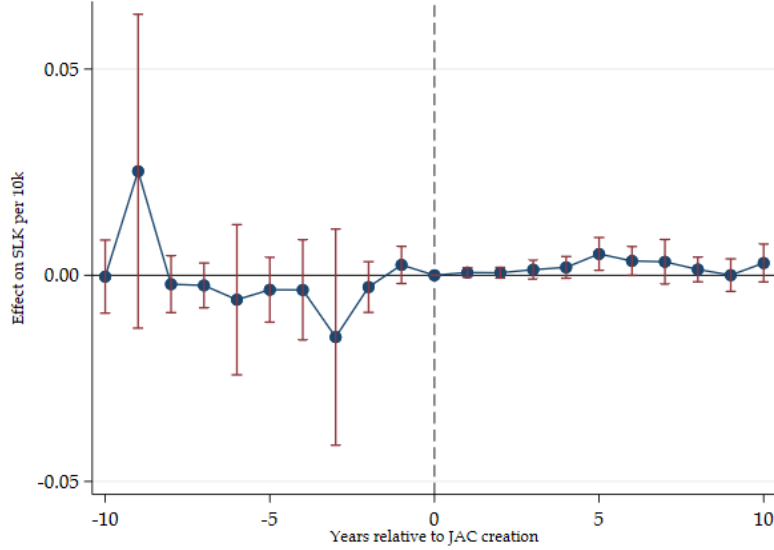


Figure H2: Dynamic DiD excluding community leaders

Notes: Dynamic event-study estimates (de Chaisemartin–D’Haultfœuille); outcome $\ln(1 + \text{SLK}/10,000)$ for social leaders *excluding líderes comunitarios* (CMH code 50); treatment new JAC per 10,000 inhabitants. Municipality and year fixed effects, department non-parametric trends, standard errors clustered at the municipality level; 95% confidence intervals. Average total effect $\hat{\beta} = 0.00163$ (SE = 0.00076), joint placebo $p = 0.536$. Source: Centro de Memoria Histórica; Ministry of Interior.

Table H7: Placebo: homicides of non-vulnerable individuals

Outcome	Av. total eff.	SE	95% CI	Placebo p
Non-vulnerable homicides	−0.00066	0.02633	[−0.052, 0.051]	0.049

Notes: Dynamic DiD (de Chaisemartin–D’Haultfœuille) with outcome $\ln(1 + \text{non-vulnerable homicides}/10,000)$, constructed from the CMH database excluding all social-leader categories. Treatment: new JAC per 10,000 inhabitants. Municipality and year fixed effects, department non-parametric trends, standard errors clustered at the municipality level. The standard error is roughly twenty-five times the baseline (Table 5); the placebo pre-trend test rejects at the 5% level. Source: Centro de Memoria Histórica; Ministry of Interior.

I Poisson (PPML)

Table I8 reports a fixed-effects Poisson (PPML) benchmark addressing the scale-sensitivity of the $\ln(1 + y)$ outcome (Chen and Roth, 2024). The outcome is the count of targeted social-leader killings and the treatment the count of new JAC registrations, with a log-population control, municipality and year fixed effects, and department-specific trends; `ppmlhdfc` drops municipalities with no killings over the panel, so the estimation sample is smaller than the full panel. Column 1 reports the contemporaneous effect; column 2 adds six annual lags of the treatment and reports the cumulative effect (the sum of the contemporaneous and lagged coefficients). The cumulative point estimate is roughly twelve times the contemporaneous one and of the same positive sign,

consistent with the dynamic build-up documented in the main event study, though the count model is considerably less precise and neither coefficient reaches conventional significance. The precise, units-robust evidence is therefore the level-outcome DiD of Table 5; the PPML serves only to confirm that the sign and the dynamic accumulation survive a units-invariant count model.

Table I8: Poisson (PPML) benchmark: count of social-leader killings

	(1) Contemporaneous	(2) Cumulative 0–6y
New JAC (count)	0.00146 (0.00878)	0.01802 (0.01972)
<i>p</i> -value	0.868	0.361
Distributed lags 1–6	No	Yes
Municipality FE	Yes	Yes
Year FE	Yes	Yes
Dept $\times$ trend	Yes	Yes
Observations	35526	33037

Notes: Fixed-effects Poisson estimated with `ppmlhdfe`. Outcome: count of targeted social-leader killings. Treatment: count of new JAC registrations. Log-population control; municipality and year fixed effects and department-specific linear trends; standard errors clustered at the municipality level. `ppmlhdfe` drops municipalities with no killings over the panel (separated observations), so N is below the full panel. Column 2 adds six annual lags of the treatment and reports the cumulative effect (sum of the contemporaneous and lagged coefficients).

J Balance on Pre-Determined Variables

Table J9 reports a formal balance test comparing conflict and non-conflict municipalities on pre-determined characteristics. Eight of the 13 variables have normalized differences exceeding the conventional $|\Delta| > 0.25$ threshold (Imbens and Rubin, 2015). The most striking imbalance is the pre-existing JAC stock: conflict municipalities averaged 19.0 JACs per 10,000 by 1980, compared to 8.7 in non-conflict municipalities ($\Delta = 0.625$). This imbalance reflects the same geographic mechanism driving the LASSO results, remote, rural municipalities attracted both early JAC adoption and, later, armed-group settlement, and does not threaten the within-municipality identification strategy, since municipality fixed effects absorb pre-existing stock levels. It does motivate using armed group character measured before 2016 as a predetermined moderator for the post-2016 outcome period, rather than relying on the full-period conflict split as a clean causal contrast.

Table J9: Balance on pre-determined variables: conflict vs. non-conflict municipalities

Variable	Mean		Balance	
	No conflict	Conflict	Norm. diff.	<i>p</i> -value
Dist. to dept. capital	86.963	79.653	-0.114	0.081
Altitude	1267.375	1087.246	-0.186	0.005
Area (km2)	863.881	1067.992	0.067	0.357
Dist. to Bogotá	348.169	312.827	-0.169	0.009
Dist. to market	168.698	117.281	-0.399 †	0.000
La Violencia (1948-53)	0.056	0.157	0.334 †	0.000
Land conflict 1901-17	0.026	0.111	0.342 †	0.000
Land conflict 1918-31	0.026	0.150	0.449 †	0.000
Spanish settlement 1510-61	0.589	0.298	-0.611 †	0.000
Indigenous pop. 1535	0.519	0.369	-0.304 †	0.000
Muni. founding date	1874.195	1869.085	-0.046	0.504
JAC stock 1980	8.676	19.024	0.625 †	0.000
<i>N</i> : 278 non-conflict, 845 conflict municipalities				

Notes: Pre-determined variables drawn from LASSO predictor set. Normalized difference with *p*-values from two-sample *t*-tests. †: $|\Delta| > 0.25$ (Imbens and Rubin, 2015).

K Robustness to the Definition of Established Armed Presence

The classification of municipalities as established-presence zones rests on a threshold of three or more active years of armed activity during 1993–2015. This threshold is motivated by Arjona (2016)’s distinction between extractive incursions and durable rebelocracy, but it is ultimately a modeling choice. This appendix tests the sensitivity of the between estimator to that choice by varying the threshold between 2 and 4 active years; the complementary specification that sidesteps the threshold by entering conflict duration continuously is reported in the main text (Section 6.3, Table 8).

Table K10 tests sensitivity to the threshold used to define established presence, comparing results for thresholds of 2+, 3+ (baseline), and 4+ active years. The positive and significant interaction for established-presence municipalities is robust across all three thresholds in the pooled specification, and the qualitative pattern in the post-2016 split is consistent across definitions. The results confirm that the heterogeneity finding does not hinge on the specific threshold chosen.

Table K10: Robustness to threshold definition of established armed presence

	Threshold: 2+ years		Threshold: 3+ years (baseline)		Threshold: 4+ years	
	Sporadic	Established	Sporadic	Established	Sporadic	Established
<i>Panel A: Pooled 1993–2024</i>						
JAC $\times$ group (pooled)	0.00090 (0.00593)	0.00914** (0.00434)	0.00049 (0.00366)	0.01014** (0.00497)	-0.00276 (0.00333)	0.01241** (0.00553)
<i>Panel B: Post-2016 only (2016–2024), JAC stock 1980 control</i>						
JAC $\times$ group (post-2016)	-0.00594 (0.00449)	0.00803 (0.00571)	-0.00518 (0.00335)	0.01050 (0.00683)	-0.00750** (0.00346)	0.01343* (0.00773)

Notes: Between estimator. Outcome: mean $\ln(1 + \text{SLK}/10,000)$ per municipality $\times$ period. Each column pair uses a different threshold to define established armed presence: 2+, 3+ (baseline), or 4+ active years during 1993–2015. Reference category: no armed presence. Panel A pools 1993–2024; Panel B restricts to post-2016 and controls for JAC stock per 10,000 in 1980. Department fixed effects throughout. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.